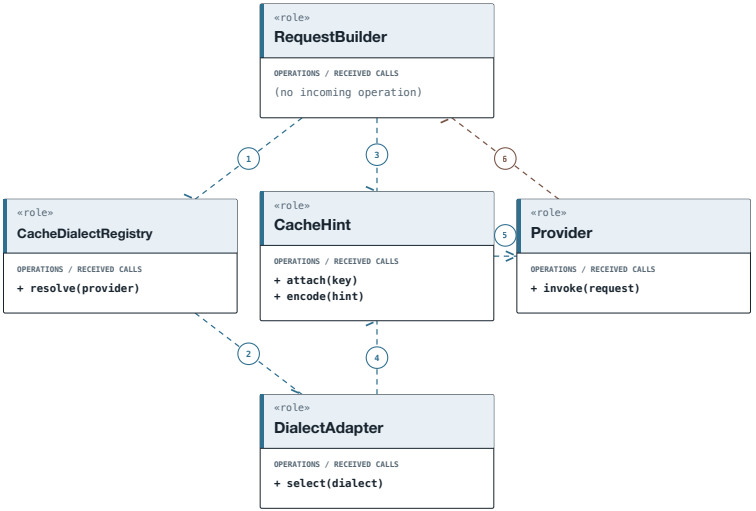
P07 Cache-dialect fanout

Primitive / UML CLASS VIEW / ANALYTICAL MODEL

Adapt caching to provider-specific interfaces.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	A request is built for a provider with cacheable context.
TRIGGER	A cache key, hint, or control record is attached.
STRUCTURE	Provider dialect adapter, cache identity, and response association.
EFFECT	Compatible cache controls are encoded and unsupported controls are rejected or omitted.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Provider capability and cache identity are known before dispatch.
ENSURES	Cache control maps to the request without changing semantic content.
INVARIANT	Cache identity never crosses user, tenant, or incompatible provider boundaries.
VIOLATION	Unsupported or invalid hints degrade explicitly without silent semantic drift.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 RequestBuilder → CacheDialectRegistry resolve(provider)
2 CacheDialectRegistry → DialectAdapter select(dialect)
3 RequestBuilder → CacheHint attach(key)
4 DialectAdapter → CacheHint encode(hint)
5 CacheHint → Provider invoke(request)
6 Provider ↩ RequestBuilder return(response)

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	Prompt cache keys, cache-control blocks, TTLs, and dialect adapters.
TEST_OBLIGATION	Test identity collisions, invalidation, unsupported providers, and privacy boundaries.
NEGATIVE_CONTROL	A generic response cache is not provider-dialect cache control.